Designing a mobile prototype for supporting financial management skills and decision making: A co-creation study with informal saving groups in Dodoma, Tanzania

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***Abstract*—Women's informal saving groups (ISGs) in Tanzania play a vital role in financial inclusion particularly to the low-income earners. However, limited financial management skills for effective financial decision-making hinder their profitabiliy and sustainability. Traditional training and operaional practice methods often fail to meet their needs due to costs, social, and mobility constraints. This study employed design science research and design thinking methodologies to co-create WanawakeApp, a mobile prototype aimed at supporting financial management and decision-making. The findings indicate that the app's enhanced financial record-keeping, loan planning, and fund management are for informed financial decision-making. Women's ISGs expressed satisfaction with the App intuitive interface and context-specific content. The study also revealed that device compatibility, sociocultural factors, and user demographics influenced skill development and app's adoption. Conversely, challenges such as connectivity, digital illiteracy, and high costs posed barriers to effective use. WanawakeApp presents a promising model for strengthening community well-being by mitigating negative impacts of financial mismanagement and created a new pathway for research and innovation using design science research (DSR) in Infomation and communication technology for develpmeng (ICT4D) to empower ISGs and informal workers.**

Keywords—Mobile learning, E-learning, Digital learning, Informal saving groups, Informal workers, financial managment

I. INTRODUCTION

Women's informal savings groups (ISGs) are community-based organizations that provide alternative financial services to address unmet needs [1] [2]. These groups assist women in saving money, accessing small loans, and obtaining insurance. Despite these benefits, women's ISGs often face many challenges related to financial management skills, which hinder informed financial decision-making. Studies have identified deficiencies in areas such as financial recording, financial planning, and fund management and control practices [3] [4] [5], that undermine decision-making processes of these groups. Financial decision-making in many women's ISGs is shaped by internal dynamics such as trust and social relationships [3] [6]. However, while these relationships foster cohesion, they often reduce accountability and lead to poor fund management. As a result, some members abscond with loans or borrow for unproductive purposes with no repayment plan, leading to default. Consequently, many groups experience cash shortages, loan recovery challenges, conflicts, bankruptcy, and even dissolution [5]. Enhancing financial management skills is therefore important to help the group to analyze group finances, mitigate risks, and effectively plan for profitability and sustainability.

Currently, skills development and operational practices for women's ISGs rely on in-person meetings, consultation, and apprenticeships. These efforts aim to build capacity in financial literacy, leadership, credit counseling, and entrepreneurship [7] [4]. However, these models face limitations, including a lack of qualified trainers, restricted mobility, and substantial social responsibilities of the female members [8]. To bridge this gap, our previous study identified mobile applications (Apps) as effective tools for enhancing financial management skills training and decision-making of women's ISGs [9].

In fact, mobile apps offer key advantages such as reduced physical interaction, lower travel expenses, increased flexibility, and timely access to critical information [9], making them suitable for delivering financial management skills training and supporting ISGs operations for effective decision-making. Several studies in ICT4D have examined the role of mobile technology in empowering informal workers across different sectors [10] [11] [12] [27]. In the context of ISGs, in Sub-Saharan Africa, successful examples of mobile technology, such as ChamaApp, SmartChama, and Digital Ledger, have shown promise in supporting savings, group coordination, financial tracking, entrepreneurship, and financial literacy [5][14][15]. Despite these benefits, a critical gap remains as many of these solutions lack contextualization

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to sufficiently address align with women's ISGs financial management training and operational practices needs, largely due to lack of end-user engagement during the design and development process. This misalignment limits both effectiveness and the sustainable adoption of these solutions.

Addressing this issue requires a structured approach, such as design science research (DSR) and design thinking, that emphasizes collaborative and iterative development through active involvement of end users. To the best of our knowledge, no prior study has applied DSR and design thinking methodologies to develop a contextualized mobile intervention to enhance financial management skills and decision-making among women's ISGs in Dodoma, Tanzania. While informal workers, such as microbusiness owners in Tanzania, have benefited from mobile tools [16], this PhD study builds on previous research by using DSR and design thinking to design, develop, demonstrate, and evaluate a mobile prototype—WanawakeApp (meaning "women" in Swahili). Mobile applications are especially relevant due to prevalent smartphone app use like WhatsApp and mobile money services [3], creating favorable conditions for the implementation of WanawakeApp. The study sought to address the following research questions:

1. How can a team of software developers and women's ISGs collaboratively co-create a contextualized mobile prototype to support financial management skills and decision-making, using DSR principles?
2. How does the co-created WanawakeApp enhance financial management skills and decision-making within women's ISGs?

II. RESEARCH APPROACH AND DESIGN

A. Design science research

The DSR approach applies scientific knowledge to design, build, and evaluate artifacts addressing real-world challenges [17], [18]. It involves four interconnected cycles: change and impact, relevance, design, and rigor cycles [19]. Author [20] further defines DSR through a six stage (problem explication, define user requirements, designing and developing artifact, demonstrating artifact and evaluating artifact) as indicated in Fig. 1. DSR's applicability spans various domains; for example, [21] developed decision-making systems, [22] applied DSR in machine learning for performance analysis, and author [23] designed mobile solution to improve interactions between microbusiness and microfinance institutions. These cases demonstrate DSR's versatility and impact in solving real-world problems across fields.

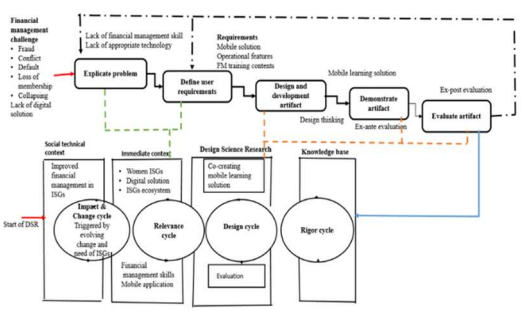


Fig. 1. A combined overview of DSR process adopted from [19], [20]

Within the DSR process, this study focused on the design cycle, which included the design, development, demonstration, and evaluation of a mobile prototype to train women's ISGs in financial management skills for decision-making. In our previous study, we utilized the relevance cycle, including problem explication, to explore smartphone use in financial management along with the challenges associated [3]. We also identified digital training requirements for women's ISGs [9], which served as acceptance criteria for evaluating the prototype's utility [17]. These DSR activities were interconnected and iterative, allowing continuous refinement of the prototype based on the feedback and evolving needs of women's ISGs.

B. Design Thinking

Design thinking emphasizes empathy, ideation, collaboration, and iterative prototyping: key elements for creating user-centered solutions [24]. Its collaborative nature encourages diverse perspectives, leading to feasible and viable contextually relevant outcomes [25]. Design thinking centers on four key guiding questions: What is it? What if? What wows? And what works?—helping designers to create more relevant and engaging solutions [26]. Like DSR, it values iteration, enabling continuous refinement to meet user needs. This study applied the design thinking principles through ideation, prototyping, and testing with women's ISGs to refine the WanawakeApp, as illustrated in Fig. 3.

C. Research design

One of the key strength of DSR is its flexibility to integrate various research methods in artifact design. We utilized both qualitative and quantitative methods throughout the development of the WanawakeApp prototype. Design thinking complemented this approach by incorporating methods such as brainstorming, interviews, mind mapping, and user testing. The process unfolded into three distinct phases as illustrated in Fig. 3.

- Ideation and brainstorming

This was a three-hour workshop session where we used brainstorming, drawing, and discussions to collect data from both women's ISGs and software developers on desired functional features as indicated in Fig. 2.

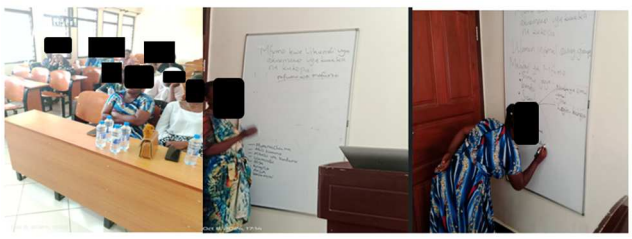


Fig. 2. Women's ISGs providing input during a brainstorming workshop.

- Design and development of WanawakeApp prototype

This phase included six-hour co-creation workshop on the second day. The team developed sample training content on financial record keeping, loan plans, use and management, loan appraisal, and saving and investment, using audio, video, and PowerPoint formats. They also created sketches to illustrate interactions with key operational features like fund management, loan processing, communication, and training. The team agreed to build the prototype as an Android app using Android Studio. Detailed design and development is discussed in the system design section.

- Evaluation of WanawakeApp prototype

Evaluation is a crucial step in DSR to ensure that the artifact meets user needs, adheres to usability criteria [27] and offers insights into the underlying design principles and theories [17]. This study evaluated women's ISG perspectives of the app using four constructs from the unified theory of acceptance and use of technology (UTAUT 1) model [28]: performance and effort expectancy, facilitating conditions, and social influence which was deemed as key factors for evaluating solution acceptance and sustained use. We purposively sampled 54 participants with relevant expertise, including 46 women's ISG members, 3 financial management experts, 2 researchers, and 3 IT experts. Participants selection was based on prior involvement and willingness to participate. Data were collected using surveys, focus group discussions (FGDs), and interviews. The surveys featured Likert scale items assessing usability, the applicability of acquired skills, and the effectiveness of operational features. The study secured research permits and ensured participant consent confidentiality. Moreover, the study ensured the protection of users sensitive information through a secure system design that integrated JSON Web Tokens for authentication and Role-Based Access Control to regulate user access. On the othr hand, financial transactions and data security in the WanawakeApp are safaguided through established financial protocols in Tanzania. Because all transaction are conducted externally via mobile money services such as Vodacom M-pesa, which are regulated by financial laws and Bank of Tanzania. This regulatory framworks minimizes the risk of fraud or cyberattacks, ensuring secure financial operations for women's ISGs

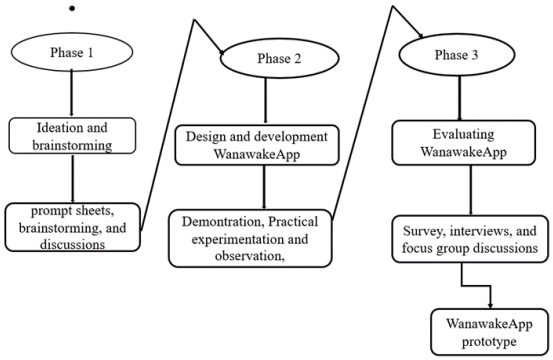


Fig. 3: Interconnected phases in the co-creation process of WanawakeApp

III. SYSTEM DESIGN FOR WANAWAKEAPP PROTOTYPE

The system design for WanawakeApp transformed the mapped functional requirements (see Table 1) into technical specifications, creating an abstract model of the app. We used unified modeling language (UML), such as use case, as detailed below for the visual representation of the app architecture and infrastructure. The prototype utilizes Flutter for its frontend development, leveraging its high performance and flexible user interface (UI) to deliver a seamless user experience, as noted by [29]. For the backend, we used the Django Python-based REST framework for robust server-side architecture. The selection of Django was driven by its security features, including JSON Web Token (JWT) to enable secure user authentication, and we implemented role-based access control (RBAC) to ensure that users can only access features and perform actions appropriate to their assigned roles and permissions. The prototype is hosted in the Linode cloud, utilizing PostgreSQL for database storage.

A. Layout design decision for WanawakeApp

- Layout design decisions include simple and linear navigation using a bottom navigation bar with three clearly labeled icons (e.g., home, training and menu) to enhance usability. The interface adopted card-based layouts displaying group members, total contributions, total loans, and total expenditure. Most importantly, buttons and icons were designed with large visual cards with minimum touch targets size of 48x48dp and generous spacing to improve accessibility partculary to low literacy users.

- Color choice: the WanawakeApp used a warm and culturally familiar palette for visual appeal and user comfort. The selected tones including deep deep blue, indigo, and green which have high contrast, improving text readability and visibility of icons**.** Moreover, colors were used to convey meanings; for example green indicate successful transaction, red failed a action or warning, and yellow/orange pending/alert.

- Text and typography: the WanawakeApp employed minimal and consice text like "pay loan" , used large, readerable fonts (16–18px for body text and 20–24px for headings). furthermore, the app used universally recognizable icons such as a dollar sign to indicate financial actions and a group of people icon presenting group members. Each icon was paired with a clear text label to enhance understanding and ensure usability across varying literacy level. The app further employed mult-language system, priotizing Kiswahili for understanding, usability and saclability.

TABLE I. FUNCTIONAL REQUIREMENTS SPECIFICATION FOR THE WANAWAKEAPP PROTOTYPE

Functionality	Descriptions
Video	Allow uploading and accessing training materials in video and audio forms
Audio	Allow uploading and accessing training materials in audio format
PDFs	Allows uploading and accessing training materials in PowerPoint and graphics format
Loan management	Enable women's ISGs apply for loans, track loan history, repay loan, leader approve or reject loans
Contribution management	Track contributions and rea-time update funds balance
Notification system	Allows sending message, make announcement, notify member about payment and call for meeting
Cordination	Allows schedule meetings, timetable and notification.
Expenditure Logging	Allows management of group expenditure

B. Quantitative performance analysis of WanawakeApp

The WanawakeApp demonstrated robust technical performance across key metric as indicated in Table II. These results highlight the system's overall stability, responsiveness and lightweight design, which contribute to higher user retention, especially users with low-end devices.

TABLE II. SYSTEM PERFOMAMNCE

Metric	Value
App Crash Rate	0.17%
API Response Time (Avg)	820ms
Server Uptime	99.98%
App Size	22 MB
language support	99.98%

C. *Use case diagram of WanawakeApp prototype*

The study employed use case diagram to identify actors such as group leaders and members, each with specific access rights within the WanawakeApp as indicated in Fig. 4.

D. *Sequential diagram of the WanawakeApp prototype*

The sequence diagram depicts the chronological flow of interactions between users (members and leaders) and the WanawakeApp for specific use cases.

E. *The main features of the WanawakeApp prototype*

The prototype runs on the Android operating system and includes several interactive features for skill development and operational practices. The prototype implements role-based access control during registration to determine user access.

The prototype workflow starts with a welcoming message and language selection (English/Kiswahili). Users then proceed to registration, and upon successful registration, users' login to access the main menu as indicated in Fig. 6.

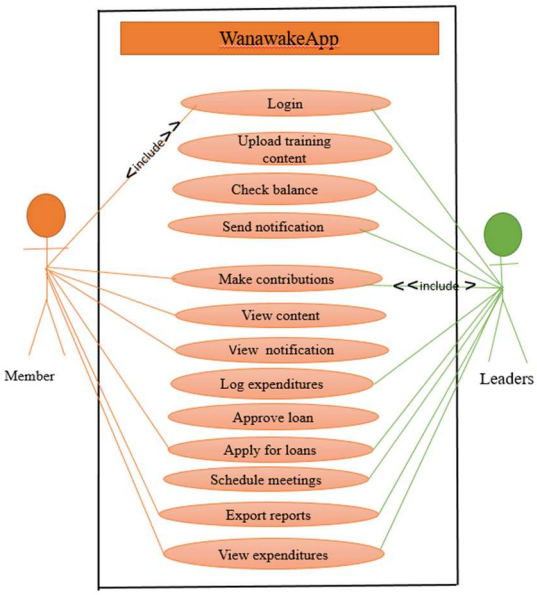


Fig. 4. Use case diagram of WanawakeApp displaying associations among actors.

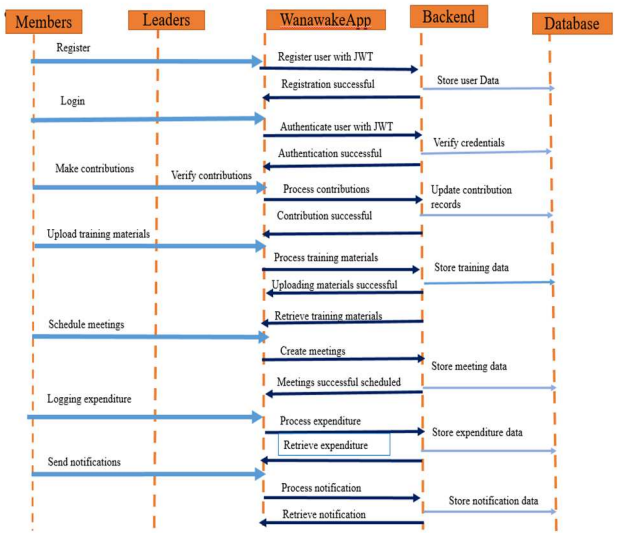


Fig. 5. Sequence diagram showing interaction among actors with the WanawakeApp prototype

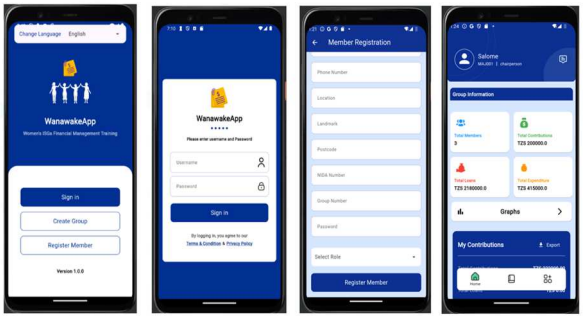


Fig. 6. Screenshot of the WanawakeApp prototype opening interface displaying user registration, login, language option and group financial status

The main menu of the regular members includes options to pay contributions and loans, view expenditure and contributions, read notifications and access training materials in various formats (see Fig.7). The leader's interface includes uploading training materials, verifying payment, reviewing and approving/rejecting loan applications, sending notifications, and logging expenditures as shown in Fig. 8.

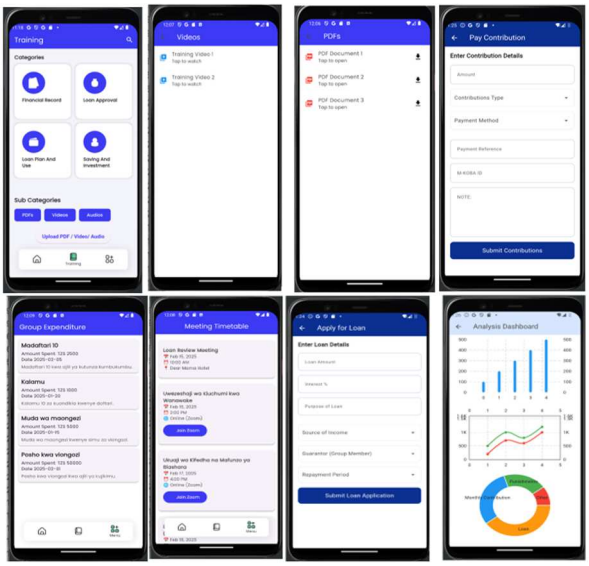


Fig. 7. Screenshots of the WanawakeApp interface for regular members, displaying features for making contributions, applying for loan, viewing expenditure, financial analytics and events as well as accessing training material.

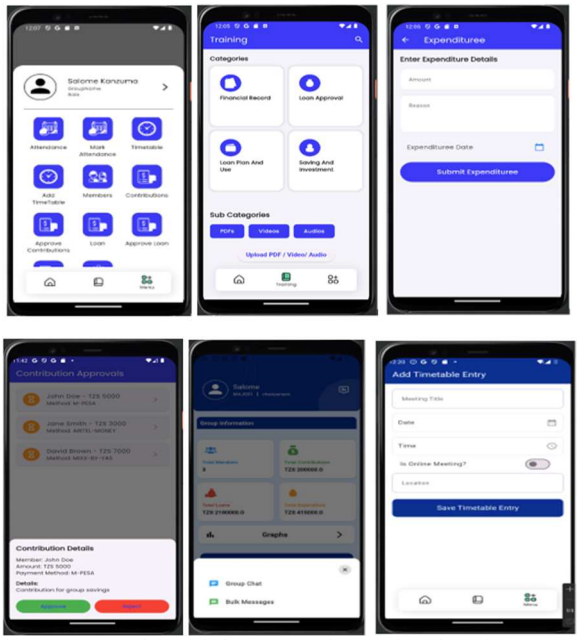


Fig. 8. Screenshots of the WanawakeApp interface for group leaders, displaying features for making contributions, logging expenditures, adding timetable, uploading training materials and approving contributions and loans

F. *Demonstrations of the WanawakeApp prototype*

Software developers and researchers introduced the WanawakeApp prototype in a three-hour workshop session to familiarize the women's ISGs with the app as indicated in Fig 9. The team guide participants through:

- Installing the APK file on their mobile devices.
- Registration and navigation
- Practiced uploading training content.
- Practiced making contributions, requesting and paying loans, and viewing expenditure.
- Practiced using messaging and video conferencing to notify, schedule, and conduct meetings

G. *Observation*

During demonstration, the team observed participants' interactions with the app to collect feedback, identify usability issues, and provide assistance. The feedback revealed both enthusiasm and challenges, especially challenge with using video conferencing, untranslated modules from English to Swahili, the need to integrate SMS system, which working separately. Participants also suggested inclusion of loan repayment analytics to visualize repayment progress and interest calculations. They further recommended including a feature for managing expenditures, which had not been included before.



Fig. 9. Screenshots showing researcher demonstrating the WanawakeApp prototype to participants.

IV. RESULTS

We conducted ex-post field evaluations of the co-created WanawakeApp prototype with ISG members and leaders using surveys, FGDs, and interviews, as discussed earlier. These evaluations, guided by the UTAUT1 model [28], assessing the app's usability, learnability, and suitability for improving financial management skills and decision-making. Participants rated performance expectancy, effort expectancy, social influence, and facilitating conditions of the app using five Likert scale items ranging from 1 (*strongly agree*) to 5 (*strongly disagree*). The results for each UTAUT 1 variable is presented in the following.

A. *Evaluation of the WanawakeApp based on performance expectancy*

The analysis of the results in Table 3 indicates a strong agreement among women's ISGs that WanawakeApp can enhance financial planning skills, improve decision-making, and simplify tasks compared to traditional methods. This result was reflected in the low mean scores—1.1905 (standard deviation [SD] = 0.39744) and 1.0714 (SD = 0.26066)—confirming these findings, participant X stated, "*the skills help to plan for the future*"; Participant F stated, *"it effective for those who fail to attend in-person training and meetings or can't use manual printings."* Participant C stated, "*The loan management section is practical, with clear steps for monitoring loans contributing to understanding our finances*."

TABLE III. PERFORMANCE EXPECTANCY OF THE WANAWAKEAPP

Statement	Performance expectance		
	N	***Mean***	***SD***
The app improved our skills in managing group finances	42	2.0476	1.03482
The content provided is relevant and applicable to our groups	42	1.9286	.83791
The app is effective in making informed financial decisions	42	1.1905	.39744
The app is more easier to complete tasks than traditional methods	42	1.0714	.26066

B. *Evaluation of the WanawakeApp prototype based on effort expectancy*

The findings in Table 4 show agreement among women's ISGs that the WanawakeApp prototype is user-friendly, ease to learn and use, supporting seamless access to training and group activities, reflecting the low mean score of 1.0476 (*SD* = 1.2476). This finding was further supported by Participant A's statement:

> *"The content is organized into, manageable units in various forms, making it easy to digest. Plus, the inclusion instructions in Kiswahili fosters our understanding".*

TABLE IV. EFFORT EXPECTANCY OF THE WANAWAKEAPP

Statement	Effort expectancy		
	N	***Mean***	***SD***
I can conveniently complete group tasks when I'm using the app	42	1.2143	1.2143
The instructions and icon arrangement are very clear and easy to follow	42	1.7619	1.7619
I can navigate through the app without any difficulty	42	1.9524	1.9524
I find it easy to use the app to learn financial management skills	42	1.0476	1.2476

C. Evaluation of the WanawakeApp prototype based on facilitating conditions

The results in Table 5 show that women's ISGs agreed with the app's facilitating conditions, particularly with the variety of resources offered, as reflected by the mean score of 2.00 (*SD* = 0.88345). However, compatibility with iOS devices emerged as an area needing improvement with a higher mean score of 2.2619 (*SD* =1.25055). This issue was reported in interviews, such as by Participant J:

> *I couldn't access the app, so I had to rely on a friend with an Android device. It worked fine on her phone, so I borrowed it to complete the tasks, but it was inconvenient.*

TABLE V. FACILITATING CONDITIONS OF THE WANAWAKEAPP

Statement	Facilitating conditions		
	N	***Mean***	***SD***
The prototype works well in my phone	42	2.2619	1.25055
My group can incorporate their operations	42	1.7619	1.10010
The prototype offers a variety of resources (function and contents) for decision making	42	2.0000	.88345
The prototype has minimal errors	42	1.9048	1.18547

D. Evaluation of the WanawakeApp prototype based on social influence

The findings in Table 6 shows that women's ISGs strongly agree with app social influence particularly promoting sustainable development and addressing misunderstanding with a means scores of 1.1905 (*SD* = 0.39744) and 1.0714 (*SD* = 0.26066) respectively. This finding translate the potential of the WanawakeApp for boosting group outcomes. Participant B shared how the app can help to prevent conflict and losses:

> *"We once suffered financial loss that led to a court case, which we lost due to missing BRELA documentation. Due to this some members left the group. Now we can track such issues and act quick using the system".*

TABLE VI. SOCIAL INFLUENCE OF THE WANAWAKEAPP

Statement	Social influence		
	N	***Mean***	***SD***
The prototype works well in my phone	42	1.1905	.39744
My group can incorporate their operations	42	1.0714	.26066
The prototype offers a variety of resources (function and contents) for decision making	42	1.5952	0.73450
The prototype has minimal errors	42	2.0476	.85404

E. Perceived challenges of using the WanawakeApp

The findings revealed digital illiteracy and unfamiliarity were barriers leading to difficulties in registration, accessing features, and using video conferencing. This made some participants feel frustrated or excluded due to repeated login issues or inability to join meetings while technical issues like poor audio, trouble unmuting, or inability to share document disrupted participation. Moreover, the results show that some members lacked interest in learning and using the app, believing that traditional financial management methods were sufficient, while age and cultural views discouraged learning new things. Furthermore, the results indicate that some women ISGs view Wanawakeapp as impersonal, preferring traditional methods for strong group cohesion. Financial constrains were also found to hinder access; high data costs and transaction fees limit regular use. Connectivity issues, voice quality, and lack of iPhone compatibility further compounded access barrier.

V. DISCUSSION OF THE RESULTS

This study co-created WanawakeApp prototype using DSR and design thinking methodologies aimed at improving financial management and decision-making in women's ISGs. The prototype featured multimedia training, video conferencing, fund management and coordination features, allowing practical skill application. The DSR change and impact cycle enhances continuous feedback, ensuring the real-world relevance of the app. The relevance cycle identified key financial management issues and shaped functional requirements, which also guided usability evaluation. The design cycle encompassed the design, development, demonstration, and evaluation of the WanawakeApp prototype while iterating with relevance and rigor cycles to align with user needs. The rigor cycle contributed to the knowledge base by offering insights into how the prototype could enhance financial management within women's ISGs. The research results are discussed based on the research questions (RQs).

A. How does the co-created WanawakeApp enhance financial management skills and decision-making within women's ISGs?

The co-creation of the WanawakeApp prototype shifted the dynamic from a top-down model to a genuine partnership by engaging end users. Although women's ISGs were hesitant at first, their confidence progressively grew due to the supportive atmosphere (see Fig. 3). These findings offer valuable insights for government and practitioners working with informal workers, stressing the importance of user-centric initiatives. The prototype proved effective in addressing self-skills development and operations for decision-making challenges, openening new avenues for research and innovation to enhance ISGs using DSR in the ICT4D arena. The transparency provided by the app can help individual members to access contribution information, total funds, expenses, loans, and repayments, thereby preventing fraud, enhances trust and accountability. However, the study experienced participant's turnover due to the fluid nature of ISG membership, which disrupted the study momentum. While such fluctuations are expected, the study advocates identifying a stable core group to ensure continuity and serve as mediators to onboard and sensitize new members to the project.

B. RQ2: How does the co-created WanawakeApp enhance financial management skills and decision-making within women's ISGs?

The evaluation of the WanawakeApp shows promise in enhancing financial management and supporting informed decision-making, particularly in loan management and information access. These findings align with existing studies

[14], [30] that highlight the role of digital solutions in addressing the socio-economic challenges of informal workers. Video conferencing in particular is a novel innovation that found to support remote coordination, reducing physical presence and proximity for both leaders and members, strengthening collective decision-making. Nevertheless, digital illiteracy remains a major barrier, causing difficulties accessing online meetings and updates, leading to missed information. This challenge is consistent with previous research, which identified digital illiteracy as a widespread issue among informal workers, often compounded by fear of monetary loss, taxation, and increased competition [30], [27], [31]. Author [32] further noted that such challenges influence technology transformation in low-income countries and highlight opportunities for ICT4D initiatives. The study recommends peer mentoring and providing clear information to dispel misconceptions to improve trust and encourage adoption.

Participants appreciated the prototype's intuitive features, simplified content, and contextualization of content into the local language (Kiswahili), which can enhance interaction, accessibility, and scalability potential. This aligns with the technological acceptance [TAM] model, which links ease of use and technology adoption [33], and the result has important implications in the lowering digital divide among ISGs and the marginalized group in Sub-Saharan Africa. However, Android inclusivity limited access for users with other operating systems, such as iOS, underscoring the need to expand Flutter's cross-platform capability and explore web-based alternatives to enhance inclusivity and reach.

The evaluation showed the prototype can enable ISGs to monitor available funds, cash flow, and assess financial performance (see Fig. 7), promoting transparency, accountability, and trust. This has the potential to mitigate harms of unsustainable debt, such familly conflicts, mental health issues, by supporting informed planning for financial stability and inclusion [34]. Improved cash flow and better loan repayments, and reduced unnecessary borrowing and spending are likely outcomes. Adhering to a financial plan cultivates discipline and saving practices essential for investment, income diversification, and settlement of loan balances [35]. Features outlining contribution and repayment terms and data analytics have the potential to reinforce application loan appraisal skills. These insights underscore the potential for integrating advanced technology like artificial intelligence and machine learning for credit scoring and fund management. Such innovation can be adopted by formal financial institutions and government loan programs to enhance loan recoverability.

Sociocultural factors such as age, education, and group history influenced skill development and app adoptions. The long-established group felt experience enough to handle financial matters, while past fraud and weakening social cohesion have fostered skepticism and distrust toward the app. Research shows social cohesion is a key drivers community and helps address lack of life insurance in the informal sector [1]. Older and less-educated members viewed skills development as irrelevant, shaping their reluctance to adopt or recommend the app, underscoring the need for inclusive design. However, poor connectivity hindered communication and updates, with some members exploiting it to delay payments [3]. High internet costs and fees further limited usage, reinforcing reliance on traditional methods. Though offline features exist, access to online functionalities remains vital. Authors [3] observed that many informal workers prioritize basic needs over airtime and data. Addressing these cost-related barriers would strengthen efforts to empower women's ISGs and the low-income earners.

C. *Limitations of the study*

The study's generalizability is limited by its focus solely on accumulating savings and credit associations (ASCAs) and rotating savings and credit associations (ROSCAs) excluding other ISGs types with potentially different needs. While the prototype showed promise in enhancing financial management skills and decision-making, its practical adoption remains uncertain and may be limited due to issues like disrupting cohesion, fears of theft, and mistrust of digital information, which may lead users to prefer traditional methods. The prototype's restriction to Android excluded iOS users, whose feedback remains unexamined.

D. *Conclusion*

The WanawakeApp prototype demonstrated how digital tools, when co-created and contextual applied, can enhance financial management skills and decision-making among women's ISGs. The study contributes to DSR in ICT4D by showcasing how technology can support self-directed skill development for informed decision-making. Ex-post evaluation using UTAUT 1 confirmed the app has the potential to improve financial recording, planning, and management for informed financial decision-making and thus promote financial inclusion and empowerment. These outcomes potentially lead to profitable investments, better repayment performance, and a reduced loan default rate, making the app a valuable tool for financial institutions and government loan programs. The prototype's intuitive interface, simplified content, and Kiswahili localization, enhance accessibility, inclusivity, and support scalability to other Kiswahili-speaking regions in Africa and have the potential to bridge the digital divide in the region.

E. *Future direction for improving ISGs and the load map to transform the prototyoe into fully development*

Upon completion of this PhD project, we intend to seek copyright permission from the University of Eastern Finland, the holder of intellectual property right of this research output. This permission will enable us to fully develop the WanawakeApp from its current prototype stage into a complete and functional mobile application that can operate in real-word context for enhancing functionality, usability and wider reach. In the future, we intend to incorporate the following features;

- To integrate the app with mobile money services, microfinance institutions, local banks, stock markets, to enable access to digital financial services including credit and diversifying investment opportunity, potential for promoting financial inclusion and economic empowerment

- To collaborate with training agencies such training institutions such as Institute of financial Management (IFM), College of Business Education(CBE),

Tanzania institute of Accountancy (TIA), credit officers, as well as community development program to facilitate delivery of financial management training through the app, ensuring easily accessible to ISGs.

- To incorporate dividend distribution and add gamified elements that reward positive financial behaviors such good payment habits, though scoring system. These features not only enhance trust but also motivation to use the app, retention of skill gained and practically applies skills in financial decision making. Additionally, we recommend to expand flutter using cross-platform, machine learning, and web-based approach.

- To integration the app with national identity verification system such as national identification authorities (NIDA) for identity verification to facilitate services such as collective health insurances, loan schemes and subsidized loans.

- To pilot the WanawakeApp with diverse forms of ISGs and mixed gander beyond initial study area to provide insights into the apps' usability and scalability across varied demographics, cultural context, economic condition and group dynamics

- The study underscores the need for formalization to ensure legal recognition and protection. Legal frameworks safeguard members against theft and fund mismanagement. Alternatively, having the ISG constitutions reviewed by legal professionals could reinforce legal compliance and operational integrity.

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